

A CYCLIC COMPLETELY QUASI-UNIFORM NESTED SQS(22)

ABSTRACT. A *nested* SQS(v) is a Steiner quadruple system together with a splitting of each block into two pairs; the *multiplicity* of a pair is the number of blocks in which it occurs as one of the two halves. The nesting is *completely quasi-uniform* if every pair of points occurs and the multiplicities are not all equal but any two differ by at most 1. Lu remarks that for $v \equiv 4 \pmod{6}$, apart from the Boolean orders and $v = 10$, no example is currently known; the smallest order left uncovered by the two results he cites is $v = 22$. No question or conjecture is attached to that remark, and this note settles nothing that anybody has posed: it simply exhibits a completely quasi-uniform nested SQS(22) invariant under the cyclic translation group $\mathbb{Z}_{22}$, printed here as 19 base nested blocks. The cyclic SQS(22) underlying it may well be classical; only the printed splitting is offered.

1. DEFINITIONS AND THE CONTEXT

Throughout, V is a v -set and $\binom{V}{2}$ its set of pairs. A *Steiner quadruple system* SQS(v) is a 3- $(v, 4, 1)$ design: a set $\mathcal{Q}$ of 4-subsets of V such that every 3-subset of V lies in exactly one member of $\mathcal{Q}$. Hanani proved that an SQS(v) exists exactly when $v \equiv 2$ or $4 \pmod{6}$ (or $v \leq 1$) [2].

A *nested* SQS(v) is a set $\mathcal{B}$ of *nested blocks* $\{x, y \mid z, w\}$, each an unordered pair of disjoint pairs, such that the underlying quadruples $\{x, y, z, w\}$ form an SQS(v); we write the two halves separated by a bar. The *multiplicity* $\mu(P)$ of $P \in \binom{V}{2}$ is the number of nested blocks having P as a half. The nesting is *uniform* if μ is constant, *quasi-uniform* if μ is *not* constant while $|\mu(P) - \mu(P')| \leq 1$ for all P, P' , and *completely* quasi-uniform if in addition $\mu(P) \geq 1$ for *every* $P \in \binom{V}{2}$. These are the definitions of Lu [3], who attributes the notion of a nested SQS to Chee, Dau, Etzion, Kiah and Zhang [1].

The remark that prompted this note is quoted from [3]:

“In contrast, for completely quasi-uniform nested SQS(v) with $v \equiv 4 \pmod{6}$, the situation remains largely unresolved. Apart from the Boolean cases and the smallest nontrivial case SQS(10) given in Example 5.1, no further examples are currently known.”

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This is a statement of the state of knowledge: Lu attaches no problem, question or conjecture to it, and names no order, so nothing below settles a posed statement. We address only $v = 22$; the orders $28, 34, 40, \dots$ are untouched, as is Problem 3 of [1], which asks for an *infinite* family. We ran no isomorphism test and searched no catalogue of cyclic quadruple systems, so we claim only that the printed object *is* a completely quasi-uniform nested SQS(22): neither the design nor its splitting is claimed to be new, and nothing here bounds the number of such systems.

The relevant orders $v \equiv 4 \pmod{6}$ with $v \geq 10$ begin $10, 16, 22, 28, \dots$. Of these, $v = 10$ is Lu's Example 5.1 and $v = 16 = 2^4$ is Boolean, covered by his theorem producing a completely quasi-uniform nested SQS(2^m) for every even $m \geq 4$. So $v = 22$ is the smallest order not covered by those two results, and it is the order treated here.

Theorem 1. *A completely quasi-uniform nested SQS(22) exists. One is invariant under the cyclic group $\mathbb{Z}_{22}$ acting by $x \mapsto x + 1$, and is the union of the orbits of the 19 base nested blocks listed in Section 2. (No claim is made that $\mathbb{Z}_{22}$ is its full automorphism group.)*

2. THE OBJECT

Let $V = \mathbb{Z}_{22} = \{0, 1, \dots, 21\}$ and let $G = \mathbb{Z}_{22}$ act by $x \mapsto x + 1 \pmod{22}$. Each of the following 19 nested blocks is listed with the length of its G -orbit; $\mathcal{B}$ is the union of those 19 orbits.

B01	{ 0, 1 2, 4}	22	B11	{ 0, 2 12, 19}	22
B02	{ 0, 1 5, 6}	22	B12	{ 0, 18 2, 6}	22
B03	{ 0, 7 1, 8}	22	B13	{ 0, 18 3, 9}	22
B04	{ 0, 1 9, 19}	22	B14	{ 0, 11 3, 14}	11
B05	{ 0, 10 1, 11}	22	B15	{ 0, 8 4, 15}	22
B06	{ 0, 14 1, 20}	22	B16	{ 0, 9 2, 7}	22
B07	{ 0, 2 8, 14}	22	B17	{ 0, 13 2, 11}	11
B08	{ 0, 3 7, 17}	22	B18	{ 0, 17 4, 9}	22
B09	{ 0, 13 3, 6}	22	B19	{ 0, 16 5, 11}	11
B10	{ 0, 5 2, 10}	22			

Explicitly, $\mathcal{B} = \{ \{a+t, b+t \mid c+t, d+t\} : \{a, b \mid c, d\} \in \{B01, \dots, B19\}, t \in \mathbb{Z}_{22} \}$, all residues mod 22.

Lemma 2. *Sixteen of the nineteen orbits have length 22 and three, those of B14, B17 and B19, have length 11, so $|\mathcal{B}| = 16 \cdot 22 + 3 \cdot 11 = 385$.*

Proof. The stabiliser of a nested block is a subgroup of $\mathbb{Z}_{22}$, so it is one of $\{0\}$, $\{0, 11\}$, the subgroup H of order 11, and $\mathbb{Z}_{22}$ itself. The orbits of H on V are its two cosets, of size 11, and $\mathbb{Z}_{22}$ is transitive on V ; since an invariant nested block has an invariant underlying quadruple, which would then be a union of orbits of size 11 or 22, neither H nor $\mathbb{Z}_{22}$ can stabilise a nested block. Hence the stabiliser is $\{0\}$ or $\{0, 11\}$, every orbit has length 22 or 11, and length 11 occurs exactly when the block is fixed by $\sigma : x \mapsto x + 11$. Now a

nested block is fixed by σ if and only if its underlying quadruple is a union of two *diameters* $\{a, a+11\}$. Indeed, a fixed nested block has a fixed quadruple, and the orbits of σ on V are exactly the eleven diameters; conversely, if the quadruple is $\{a, a+11, b, b+11\}$ then σ preserves each of its three splittings, fixing both halves of $\{a, a+11\} \mid \{b, b+11\}$ and interchanging the halves of the other two. Among $B01, \dots, B19$ the quadruples that are unions of diameters are exactly

$$B14 : \{0, 11\} \cup \{3, 14\}, \quad B17 : \{0, 11\} \cup \{2, 13\}, \quad B19 : \{0, 11\} \cup \{5, 16\};$$

hence three orbits of length 11 and sixteen of length 22. $\square$

3. THE COUNTS

Proof of Theorem 1. Everything is a finite count on 22 points, carried out on the 385 nested blocks obtained from the table of Section 2.

It is an SQS(22). The 385 nested blocks have 385 distinct underlying quadruples, and $385 = \frac{22 \cdot 21 \cdot 20}{24}$; each of the $\binom{22}{3} = 1540$ triples lies in exactly one of them. (Equivalently, and as a second count: each of the 231 pairs lies in exactly $\frac{v-2}{2} = 10$ of the quadruples, and each point in $\frac{21 \cdot 20}{6} = 70$.)

It is nested and complete. Every block is split into two disjoint pairs by construction, and all $\binom{22}{2} = 231$ pairs occur as a half.

It is quasi-uniform, not uniform. The multiplicity profile is

$$\#\{P : \mu(P) = 3\} = 154, \quad \#\{P : \mu(P) = 4\} = 77,$$

two values differing by 1 and not all equal, with $154 \cdot 3 + 77 \cdot 4 = 770 = 2 \cdot 385$.

It is $\mathbb{Z}_{22}$ -invariant because $\mathcal{B}$ is by definition a union of $\mathbb{Z}_{22}$ -orbits. $\square$

Remark 3. Chee, Dau, Etzion, Kiah and Zhang [1] use the same terminology without the non-uniformity clause, i.e. for them “quasi-uniform” only requires multiplicities differing by at most 1. The object above is quasi-uniform in both senses, since its multiplicities are 3 and 4.

4. VERIFICATION

Every claim of Theorem 1 is a finite count on the 385 nested blocks generated by the table of Section 2, and a reader who expands that table needs nothing else: the 19 base blocks, the rule $x \mapsto x+1$, and a tally of triples and of pair multiplicities. A short standard-library Python program, `verify.py`, accompanies this note; it reads the 19 lines exactly as printed above and checks their orbit lengths (Lemma 2), the 385 distinct underlying quadruples, the unique coverage of all 1540 triples, completeness, the multiplicity profile $\{3^{154}, 4^{77}\}$, and invariance under $x \mapsto x+1$. Its recorded run is in `verify.output.txt`. The quotation and definitions above were transcribed by hand and are not machine-checked.

REFERENCES

- [1] Y. M. Chee, S. H. Dau, T. Etzion, H. M. Kiah, and X. Zhang, *Pairs in nested Steiner quadruple systems*, J. Combin. Des. **33** (2025), no. 5, 177–187. doi:10.1002/jcd.21973; arXiv:2410.14417.
- [2] H. Hanani, *On quadruple systems*, Canad. J. Math. **12** (1960), 145–157. doi:10.4153/CJM-1960-013-3.
- [3] X.-N. Lu, *Completely (quasi-)uniform nested Boolean Steiner quadruple systems*, J. Combin. Des. **34** (2026), no. 3, 139–156. doi:10.1002/jcd.22016; arXiv:2509.06663v3. The quotation above is from the final paragraph of Section 5 of arXiv:2509.06663v3; section and example numbers refer to that version, and the journal numbering may differ.